排队论

M/M/1 Kendall 符号

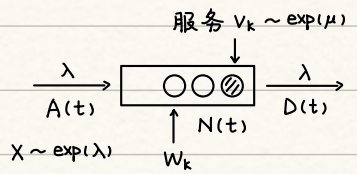
到达过程 $A(t)$ 服从 Poisson 分布 $\text{Poisson}(\lambda)$

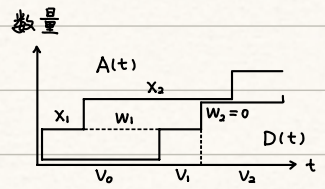
到达相隔时间 $X_k \sim \exp(\lambda)$

服务时间 V_k 服从指数分布 $\exp(\mu)$

等待时间 W_k

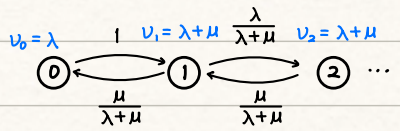
Burke 定理：若 $\lambda < \mu$，则离开过程服从 Poisson 分布，$\text{Poisson}(\lambda)$



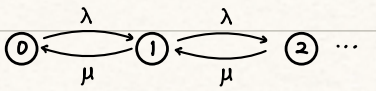


令 $\rho = \frac{\lambda}{\mu} < 1$ 稳态队列长度 $P\{N(t)=n\} = \rho^n(1-\rho)$

嵌入 Markov 链 转移概率



可数态 Markov 过程 转移率

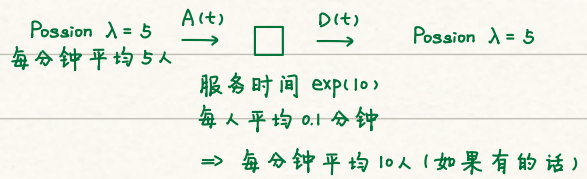


平均服务时间 $\bar{V} = \frac{1}{\mu}$

平均等待时间 $\bar{W} = \frac{1}{\mu-\lambda}$

平均排队长度 $\bar{L} = \frac{\lambda}{\mu-\lambda} = \frac{\rho}{1-\rho}$

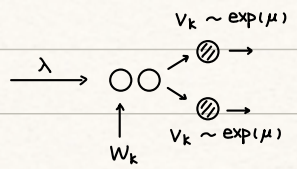
Little's 定理 $\bar{L} = \lambda\bar{W}$



M/M/2

$V = \min\{V_1, V_2\} \sim \exp(2\mu)$

$$\bar{W} = \frac{\frac{1}{\mu}}{1-\frac{\lambda^2}{4\mu^2}}, \quad \bar{L} = \frac{\frac{\lambda}{\mu}}{1-\frac{\lambda^2}{4\mu^2}}$$



$\lambda < 2\mu$ 时稳定

M/M/1/K

排队长度最长 K

$$P\{N(t)=n\} = \frac{\rho^n(1-\rho)}{\ldots^{K+1}}$$

$1 - \rho^{\ldots}$